* Template Matching

→ Slide template Image (T) over a target image (I) and use a similarity / dissimilarity function at each point.

→ If the Score is good enough it is a match

+ Similarity / Dissimilarity Functions

→ let I be the target image and T be the template image of shape m x n and patch $I_{i,j}$ be the region of I starting at (i, j)

1) SSD (Sum of Squared Difference)

$$SSD(i,j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} [I(i+m, j+n) - T(m,n)]^2$$

→ square the pixel by pixel difference

→ Fast

→ Not intensity variant (brightness affect results)

2) SAD (Sum of Absolute Difference)

$$SAD(i,j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} |I(i+m, j+n) - T(m,n)|$$

→ take absolute Value of pixel difference

→ Not intensity Variant.

3) NCC (Normalized Cross-Correlation)

$$NCC(i,j) = \frac{\sum I(i+m, j+n) \cdot T(m,n)}{\sqrt{\sum I(i+m, j+n)^2} \cdot \sqrt{\sum T(m,n)^2}}$$

→ Cosine similarity b/w template & patch (flattened as vectors).

→ Invarient to linear intensity change (contrast)

→ Not invariant to additive bias (brightness shift)

4) ZNCC (Zero-mean Normalized Cross-correlation)

subtract mean before applying NCC:-

$$ZNCC(i,j) = \frac{\sum [I(i+m, j+n) - \mu(I_{ij})] \cdot [T(m,n) - \mu(T)]}{\sqrt{\sum [I(i+m, j+n) - \mu(I_{ij})]^2} \cdot \sqrt{\sum [T(m,n) - \mu(T)]^2}}$$

→ Invariant to affine intensity changes (contrast + brightness)

* Shape-Based Matching

→ Edge-base template matching that works as follows

① Use and edge detector to extract a set of control points $\{P_1 \dots P_n\}$ in the template image T.

② Compute the gradient and normalizedit to unit vector at each point P_K:

$$\nabla T(P_K) = \begin{pmatrix} \partial_x T(P_K) \\ \partial_y T(P_K) \end{pmatrix} \qquad u_K(P_K) = \frac{\nabla T(P_K)}{\|\nabla T(P_K)\|}$$

③ Given a patch I_{ij} of the target image.

→ Compute the gradient and normalized it to unit vector at the point $\{P_1(i,y) \dots, P_n(i,y)\}$ corresponding to the control points of the template image.

$$\nabla I_{ij}(P_K) = \begin{pmatrix} \partial_x I_{ij}(P_K) \\ \partial_y I_{ij}(P_K) \end{pmatrix} \qquad u_K(P_K) = \frac{\nabla I_{ij}(P_K)}{\|\nabla I_{ij}(P_K)\|}$$

4) Compute the similarity as the mean cosine similarity of each pair of gradient.

$$s(T, I_{ij}) = \frac{1}{n} \sum_{k=1}^{n} u_k(p_k) \cdot u_k(p_k) = \frac{1}{n} \sum_{k=1}^{n} \cos\theta_k \in$$

$[-1, 1]$

$s(T, I_{ij}) = 1$

→ when the gradients perfectly match

→ A minimum threshold s_{min} is used to determine if there is a match.

* Invariance to contrast polarity inversion

→ An object might appear on a darker or brighter background, more robust similarity function can be employed

* Global contrast polarity inversion

$$s(i,j) = \frac{1}{n} \left| \sum_{k=1}^{n} u_k(p_k) \cdot \tilde{u}_k(\tilde{p}_k) \right|$$

$$= \frac{1}{n} \left| \sum_{k=1}^{n} \cos\theta_k \right|$$

* local contrast polarity function

$$S(ij) = \frac{1}{n} \sum_{k=1}^{n} \left| u_k(p_k) \cdot u_k(p_k) \right|$$

$$= \frac{1}{n} \sum_{k=1}^{n} \left| \cos\theta_k \right| \longrightarrow$$ More robust one.